

Frozen-Jacobian Line Search for the Geodesic IPM

1 Setup

We consider the geodesic interior-point method for the conic program

$$\min_x c^T x \quad \text{s.t.} \quad Ax + b \succeq 0,$$

with the weight parameterization

$$s = \frac{1}{k} W^{-1}(e - d), \quad \lambda = \frac{1}{k} W(e + d),$$

where $k = 1/\sqrt{\mu}$ and W is a positive diagonal scaling (or, more generally, an element of the automorphism group of the cone). We write $P(W) = W^2$ for the quadratic representation.

Eliminating d from the KKT conditions $s = Ax + b$ and $A^T \lambda = c$ yields the normal equations

$$(A^T W^2 A + Q) y = 2A^T W - k(c + A^T W^2 b), \tag{1}$$

where $y = kx$ and Q accounts for any quadratic cost. The Newton direction is recovered as $d = e + W$ slack, where $\text{slack} = -kb - Ay$.

The standard geodesic LP algorithm repeats:

1. Factor $G = A^T W^2 A + Q$.
2. Decompose into $d = d_0 + k d_1$ (2 back-solves).
3. Line search: find $k_{\max}$ with $\|d_0 + k_{\max} d_1\|_{\infty} \leq 1$.
4. Geodesic step: $W \leftarrow W \cdot \exp(\alpha d)$.

Each outer iteration requires one factorization and two back-solves.

2 Frozen-Jacobian Newton system

When we step from W_0 (the factorization point) to a new iterate W_i , the standard approach refactors at W_i . Instead, we freeze the Jacobian at W_0 and only update the residual at W_i .

The standard linearized equations at W with direction d are:

$$A^T(W(e+d))/k = Qx + c, \quad (2)$$

$$W^{-1}(e-d)/k = Ax + b. \quad (3)$$

In the frozen-Jacobian variant, the “value” terms use W_i (current iterate) while the “Jacobian” terms use W_0 (frozen):

$$A^T(W_i + W_0 d)/k = Qx + c, \quad (4)$$

$$(W_i^{-1} - W_0^{-1} d)/k = Ax + b. \quad (5)$$

2.1 Elimination

From (5):

$$d = W_0(W_i^{-1} - k(Ax + b)) = W_0W_i^{-1} - k W_0(Ax + b).$$

Substituting into (4):

$$A^T(W_i + W_0^2W_i^{-1} - k W_0^2(Ax + b))/k = Qx + c.$$

Rearranging:

$$\boxed{(A^T W_0^2 A + Q) x = \frac{1}{k} A^T(W_i + W_0^2 W_i^{-1}) - A^T W_0^2 b - c.} \quad (6)$$

Consistency check. When $W_i = W_0$, the centering term becomes $W_0 + W_0^2 W_0^{-1} = 2W_0$, recovering (1).

2.2 Direction recovery

After solving (6) for $y = kx$:

$$d = \frac{W_0}{W_i} + W_0 \cdot \text{slack}, \quad \text{slack} = -kb - Ay. \quad (7)$$

Compare with the standard recovery $d = e + W \cdot \text{slack}$: the centering term changes from e to W_0/W_i , and the Jacobian in the slack term changes from W_i to W_0 .

3 Two-term decomposition

The frozen system (6) decomposes into centering and cost components:

$$(A^T W_0^2 A + Q) y_0 = A^T(W_i + W_0^2 W_i^{-1}), \quad (\text{centering, depends on } W_i) \quad (8)$$

$$(A^T W_0^2 A + Q) y_1 = -(c + A^T W_0^2 b) + d_{\text{eq}}, \quad (\text{cost, frozen}) \quad (9)$$

where d_{eq} is the equality constraint correction.

Key observation. The cost component y_1 depends only on W_0 and is *constant* across all frozen-Jacobian iterations. After the initial 2-solve decomposition, each subsequent iteration requires only **one back-solve** (for y_0) plus a line search on $d = d_0^{(i)} + k d_1^{\text{frozen}}$.

The directions are recovered componentwise:

$$d_0 = W_0 \cdot (W_i^{-1} - A y_0), \quad (10)$$

$$d_1 = W_0 \cdot (-b - A y_1). \quad (\text{frozen}) \quad (11)$$

4 Algorithm

1. **Factor:** Set $W_0 \leftarrow W_i$. Compute $G = A^T W_0^2 A + Q$ and factor. Decompose $d = d_0 + k d_1$ via standard 2-solve decomposition. Line search for k . Take geodesic step.
2. **Frozen-Jacobian iterations** (repeat until degradation):
 - (a) Compute $d_0^{(i)}$ via 1 back-solve with RHS from (8). Reuse d_1^{frozen} from step 1.
 - (b) Line search: find $k_{\max}$ with $\|d_0^{(i)} + k_{\max} d_1^{\text{frozen}}\|_{\infty} \leq 1$. Set $k \leftarrow \max(k, k_{\max})$.
 - (c) Take geodesic step $W_i \leftarrow W_i \cdot \exp(\alpha d)$.
 - (d) If $\|d_0^{(i)}\|$ is growing (frozen Jacobian degrading), exit.
3. Go to step 1 (refactor at new W_i).

Cost per outer iteration. One factorization, $2 + n_{\text{inner}}$ back-solves (2 for the initial decomposition, 1 per frozen-Jacobian step), and $1 + n_{\text{inner}}$ line searches.

5 Results

On a test QP (HS21-type, 7 variables, 10 constraints):

Method	Factorizations	Back-solves	Final gap
Standard GeodesicLP	16	32	9.7×10^{-11}
Frozen-Jacobian ($n_{\text{inner}} = 3$)	9	50	8.4×10^{-12}

The frozen-Jacobian variant halves the number of factorizations at the cost of additional back-solves. For problems where factorization dominates (large sparse systems), this is a significant win.

5.1 Convergence behavior

The gap per factorization shows accelerating convergence:

Fac	k	Gap	k ratio
0	1.7	2.8	—
1	4.1	6.5×10^{-1}	$2.4\times$
2	8.2	4.2×10^{-2}	$2.0\times$
3	30	6.0×10^{-3}	$3.7\times$
4	136	4.0×10^{-4}	$4.5\times$
5	577	1.2×10^{-5}	$4.2\times$
6	7.3×10^3	3.0×10^{-8}	$12.7\times$
7	4.4×10^4	1.8×10^{-9}	$6.0\times$
8	7.5×10^5	8.4×10^{-12}	$17.2\times$

The k -ratio (hence μ -reduction) per factorization accelerates as the algorithm approaches the solution. Near the optimum, d_1 is small (the cost direction vanishes), so the frozen Jacobian stays accurate over many steps and the line search can push k much further before the approximation degrades. This produces near-quadratic convergence in the number of factorizations.

5.2 Why frozen-Jacobian centering alone fails

An earlier experiment froze the Jacobian but held k fixed (pure centering, no line search). This reduced $\|d\|_\infty$ from 1.0 to ~ 0.6 per step, but the subsequent fresh decomposition showed the same $\|d_0\| \approx 1$ regardless. The centering did not help the next line search because d_0 (the centering residual at a fresh factorization) is determined by the distance from W to the central path at the *current* k —not by how well centered W was at the *previous* k .

The crucial ingredient is combining the frozen-Jacobian with a line search: this advances k (reduces μ) using only cheap back-solves, avoiding the factorization cost of the standard approach.

6 Extension to θ -continuation

The θ -continuation algorithm (see `geodesic_newton_direction.tex`) uses a three-term decomposition

$$d(k, \tau, \theta) = d_0 + k(\tau d_1^{(0)} + \theta d_1^{(\theta)})$$

and selects (τ, θ) jointly via the duality identity (gap equation) and a normalization equation. Here we derive the frozen-Jacobian variant.

6.1 Frozen-Jacobian primal-dual equations

At frozen Jacobian W_0 and current iterate W_i , the θ -blended primal-dual equations are:

$$\lambda = \frac{1}{k}(W_i + P(W_0^{1/2})(d)), \quad (12)$$

$$s = \frac{1}{k}(W_i^{-1} - P(W_0^{-1/2})(d)), \quad (13)$$

with primal feasibility $s = Ax + \tau b + \theta(e - b)$ and dual stationarity $A^T \lambda = \tau c + \theta(A^T e - c) + d_{\text{eq}}$.

Comparing with the standard parameterization where $\lambda = \frac{1}{k}P(W^{1/2})(e + d)$: the *value* W_i replaces $P(W^{1/2})(e)$, and the *Jacobian* $P(W_0^{1/2})$ replaces $P(W^{1/2})$.

6.2 Three-term decomposition

Eliminating d from (13) as in Section 2.1, the Gram system $(A^T P(W_0)A + Q)y = \text{rhs}$ has three right-hand sides:

$$\text{rhs}_0 = A^T(W_i + P(W_0)(W_i^{-1})), \quad (\text{centering, depends on } W_i) \quad (14)$$

$$\text{rhs}_1 = -(c + A^T P(W_0)b) + d_{\text{eq}}, \quad (\text{cost, frozen}) \quad (15)$$

$$\text{rhs}_2 = -\text{rhs}_1 - A^T(e + P(W_0)e), \quad (\theta \text{ correction, frozen}) \quad (16)$$

yielding solutions $y_0, y_1^{(0)}, y_1^{(\theta)}$ with $y = y_0/k + \tau y_1^{(0)} + \theta y_1^{(\theta)}$.

The directions are recovered using $P(W_0^{1/2})$:

$$d_0 = P(W_0^{1/2})(W_i^{-1} - Ay_0), \quad (17)$$

$$d_1^{(0)} = P(W_0^{1/2})(-b - Ay_1^{(0)}), \quad (\text{frozen}) \quad (18)$$

$$d_1^{(\theta)} = P(W_0^{1/2})(b - e - Ay_1^{(\theta)}). \quad (\text{frozen}) \quad (19)$$

Key observation. Only rhs_0 (and hence d_0) depends on W_i . The cost and theta components $d_1^{(0)}, d_1^{(\theta)}$ are frozen. After the initial 3-solve decomposition, each frozen-Jacobian step requires **one back-solve** to refresh d_0 .

Consistency check. When $W_i = W_0$: $P(W_0)(W_0^{-1}) = e$, so $\text{rhs}_0 = A^T(W_0 + e)$...no, that is wrong. We have $W_0 + P(W_0)(W_0^{-1}) = W_0 + W_0 = 2W_0$, recovering the standard $\text{rhs}_0 = A^T(2W)$. And $P(W_0^{1/2})(W_0^{-1}) = W_0 \cdot W_0^{-1} = e$, recovering $d_0 = e - P(W_0^{1/2})(Ay_0)$. ✓

6.3 Duality identity with frozen Jacobian

The duality identity (gap equation) from `geodesic_newton_direction.tex` §6 is

$$b^T \lambda + c^T x + d^T \nu + \frac{x^T Q x}{\tau} + \frac{\mu}{\tau} = \theta R, \quad (20)$$

where $R = \langle b, e \rangle + 1$. With the frozen Jacobian, λ and x are:

$$\lambda = \frac{1}{k} (W_i + P(W_0^{1/2})(d)) = \frac{W_i}{k} + \tau P(W_0^{1/2}) d_1^{(0)} + \theta P(W_0^{1/2}) d_1^{(\theta)} + \frac{1}{k} P(W_0^{1/2})(d_0), \quad (21)$$

$$x = \frac{y_0}{k} + \tau y_1^{(0)} + \theta y_1^{(\theta)}. \quad (22)$$

Both are affine in (τ, θ) . The key difference from the standard case is in the $b^T \lambda$ term: the constant part is $b^T W_i / k$ (depends on W_i) rather than $b^T P(W^{1/2})(e + d_0) / k$ (depends on a single W).

Decomposing $b^T \lambda$.

$$b^T \lambda = \underbrace{\frac{1}{k} (b^T W_i + b^T P(W_0^{1/2})(d_0))}_{\sigma_0} + \underbrace{b^T P(W_0^{1/2}) d_1^{(0)}}_{\sigma_1} \tau + \underbrace{b^T P(W_0^{1/2}) d_1^{(\theta)}}_{\sigma_\theta} \theta. \quad (23)$$

Compare with the standard case where $\sigma_0 = \frac{1}{k} b^T P(W^{1/2})(e + d_0)$: the frozen version splits the “value” W_i from the “Jacobian” $P(W_0^{1/2})$.

The σ_1 and σ_θ terms depend only on W_0 (frozen). Only σ_0 changes when W_i is updated.

The $c^T x$ and $d^T \nu$ terms. These use $y_0, y_1^{(0)}, y_1^{(\theta)}$ from the frozen Gram solve:

$$c^T x = \frac{c^T y_0}{k} + c^T y_1^{(0)} \tau + c^T y_1^{(\theta)} \theta = \frac{\gamma_0^{(c)}}{k} + \gamma_1 \tau + \gamma_\theta \theta, \quad (24)$$

$$d^T \nu \text{ enters through } d_{\text{eq}} \text{ in the same pattern.} \quad (25)$$

The γ_1 and γ_θ terms use the frozen $y_1^{(0)}$ and $y_1^{(\theta)}$. Only $\gamma_0^{(c)} = c^T y_0$ changes when d_0 is refreshed.

Quadratic cost. The $x^T Q x / \tau$ term decomposes identically to the standard case (see `geodesic_newton_direction.tex` §6.2), since $x = y_0 / k + \tau y_1^{(0)} + \theta y_1^{(\theta)}$ has the same structure. The frozen Jacobian only affects how y_0 is computed, not the algebraic form.

6.4 Violation quadratic

Following the same elimination as the standard case (`geodesic_newton_direction.tex` §8), we use the normalization equation to express $\theta(\tau) = a_0 + a_1 \tau$, substitute into the gap equation, and obtain

$$\beta \tau^2 + \gamma \tau + \mu_{\text{eff}} = 0, \quad (26)$$

where the coefficients have the same algebraic form as the standard case but with:

- σ_0 computed from W_i and W_0 via (23),
- σ_1, σ_θ computed from W_0 only (frozen),
- y_0 from the frozen Gram solve (refreshed), $y_1^{(0)}, y_1^{(\theta)}$ frozen.

What changes per frozen-Jacobian step. After refreshing d_0 and y_0 (1 back-solve):

1. Recompute $\sigma_0 = \frac{1}{k}(b^T W_i + b^T P(W_0^{1/2})(d_0))$ and $\gamma_0^{(c)} = c^T y_0$.
2. Recompute the quadratic cost terms involving y_0 : $f = y_0/k + \theta y_1^{(\theta)}$, $f^T Q f$, $f^T Q y_1^{(0)}$.
3. Solve the quadratic (26) for τ , recover θ .
4. Binary search over θ (or direct selection) as in the standard algorithm.

The normalization coefficients N_τ, N_θ also have a W_i -dependent part (through λ_0) that must be refreshed.

6.5 Algorithm for frozen-Jacobian θ -continuation

1. **Factor:** Set $W_0 \leftarrow W_i$. Compute $G = A^T P(W_0)A + Q$ and factor. Decompose into $(d_0, d_1^{(0)}, d_1^{(\theta)})$: 3 back-solves. Select (τ, θ) via gap + normalization. Take geodesic step.
2. **Frozen-Jacobian step:**
 - (a) Refresh d_0 via 1 back-solve with $\text{rhs}_0 = A^T(W_i + P(W_0)(W_i^{-1}))$.
 - (b) Recompute the W_i -dependent duality coefficients (σ_0 , normalization constant, quadratic cost terms involving y_0).
 - (c) Select (τ, θ) from the updated quadratic.
 - (d) Take geodesic step.
3. Go to step 1 (refactor at new W_i).

Cost per outer iteration. One factorization, $3 + n_{\text{inner}}$ back-solves (3 for the initial decomposition, 1 per frozen-Jacobian step), and $1 + n_{\text{inner}}$ (τ, θ) selections. The frozen components $(d_1^{(0)}, d_1^{(\theta)}, \sigma_1, \sigma_\theta, \gamma_1, \gamma_\theta, \text{normalization coefficients } N_\tau, N_\theta)$ are computed once and reused.